

NEUROSTATE: GRADIENT-ISOLATED BOUNDARY LEARNING FOR INTERPRETABLE EEG TRANSFORMERS

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ABSTRACT

Transformer-based electroencephalography (EEG) classifiers achieve strong accuracy but offer no structural account of *when* brain states change, which limits their use where transition timing guides clinical decisions. We introduce NeuroState, a multi-resolution EEG transformer whose auxiliary boundary module estimates changepoint probabilities that condition attention. Our central result is an optimization one. Across eleven controlled interventions we document *auxiliary head collapse*, a failure mode in which jointly trained boundary modules converge to near-constant outputs, and we show that the deciding factor is not the loss or the mask but the order in which gradients are allowed to flow. Attention-Contrastive Boundary Learning (ACBL) with gradient isolation, a three-phase curriculum that detaches boundary gradients from the classification path during a formation phase, is the only one of the eleven that yields non-degenerate boundaries. Over five seeds NeuroState attains Cohen’s $\kappa = 0.596 \pm 0.018$ on Sleep-EDF staging and area under the receiver operating characteristic curve (AUROC) 0.924 ± 0.014 on CHB-MIT cross-subject seizure detection, with AUROC 0.913 on TUAB. A shuffled-input null shows that the Sleep-EDF boundaries carry measurable content in every seed, while conditional tests on CHB-MIT find no transition-related structure, so we present the accompanying concept-based and spectral analyses as preliminary and scope our conclusions to the EEG settings measured here.

Index Terms— EEG transformer, changepoint detection, auxiliary head collapse, gradient isolation, sleep staging

1. INTRODUCTION

Electroencephalography (EEG) signals exhibit temporally structured brain states, such as sleep stages and ictal episodes, separated by abrupt regime boundaries [1]. The timing of these boundaries carries clinical weight: seizure onset latency informs intervention decisions [2], and sleep stage transition patterns inform the diagnosis of disorders such as narcolepsy and sleep apnea [3]. An EEG model that classifies accurately but cannot localize transitions is therefore providing only an incomplete clinical picture.

Deep learning has advanced EEG classification considerably. AttnSleep [4] couples a multi-resolution convolutional encoder with adaptive feature recalibration; SleepTransformer [5] adds attention heatmaps with uncertainty quantification (UQ); L-SeqSleepNet [6] folds 200-epoch sequences into intra- and inter-subsequence attention; and foundation models such as the Biosignal Transformer (BIOT) [7] and the Large Brain Model (LaBraM) [8] generalize across tasks through self-supervised pretraining. All apply uniform self-attention across temporal positions, with no mechanism separating within-state dynamics from transition signatures. Classical

changepoint methods such as Pruned Exact Linear Time (PELT) [9] give sharp boundaries but are disconnected preprocessing stages that cannot be jointly optimized. Outside EEG, auxiliary boundary detection improves temporal action segmentation [10], yet the optimization dynamics of such heads inside transformers remain uncharacterized, and shortcut learning [11] warns that adding a head does not guarantee meaningful structure.

This paper makes two contributions. First, we characterize *auxiliary head collapse*, the systematic failure of auxiliary boundary modules inside transformers, through eleven controlled experiments spanning five categories of intervention. Second, we propose *gradient isolation*, a three-phase curriculum with a directional stop-gradient during a formation phase, which is the only one of the eleven that yields non-degenerate boundaries. Section 2 formalizes the two mechanisms behind the collapse, Section 3 describes the architecture and curriculum, and Section 4 reports the sweep together with a calibrated account of how far the resulting boundaries can be interpreted.

2. PROBLEM FORMULATION

We seek a classifier that jointly produces a class label and a boundary probability sequence $\mathbf{b} \in [0, 1]^L$ over L tokens, where $\mathbf{b}$ enters through a content-dependent mask $M_{ij}^{(h)}(\mathbf{b})$ modulating the pre-softmax attention logits $a_{ij}^{(h)}$ in head h . Sub-epoch changepoint annotations do not exist, so the boundary head must be supervised indirectly: through representation contrast, Gaussian targets at known label transitions, and a structural prior on the attention pattern.

Joint optimization of $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{aux}}$ admits two pathologies. A *shortcut* pathology arises when $M_{ij}^{(h)}(\mathbf{b})$ yields useful structure even for uninformative $\mathbf{b}$, so the classifier never requires the boundary parameters θ_{bnd} to develop content [11]. A *gradient-interference* pathology arises when classification gradients move encoder representations in directions that make the boundary head’s target recede. Both drive the head toward flat distributions, and both point to the same remedy: if the shortcut exists only while the two objectives share a gradient path, separating those paths for part of training should remove it. Section 3.2 builds a curriculum on that idea and Section 4.2 tests it against ten alternatives.

3. NEUROSTATE ARCHITECTURE AND ACBL

3.1. Architecture

Fig. 1 summarizes the architecture. Each 30-second EEG epoch is encoded at three temporal resolutions (100, 50, 25 Hz), each branch containing a temporal convolution, a spatial convolution, average

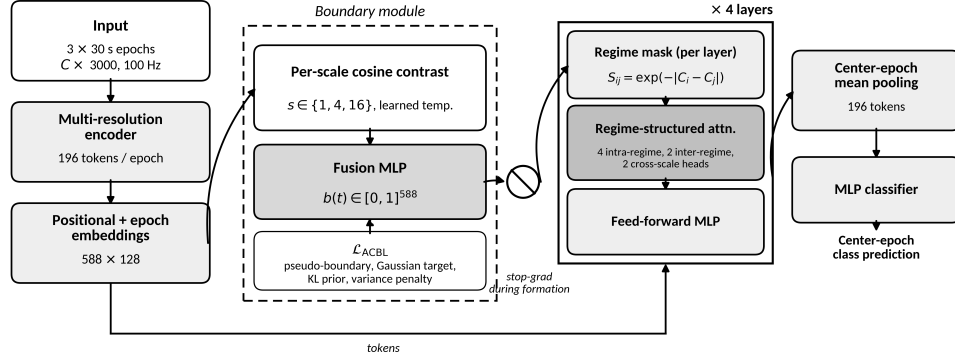


Fig. 1. NeuroState. During the formation phase a directional stop-gradient blocks the classification gradient from reaching the boundary head while the ACBL losses continue to update it; this is the mechanism that prevents auxiliary head collapse. The regime mask is rebuilt from $\mathbf{b}(t)$ in every attention layer, and classification reads the center epoch only.

pooling and a linear projection. The branches are interpolated to a common length of 196 tokens and merged by a learned linear projection, capturing both fast transients such as spindles and K-complexes [3] and slow structural drift. The three-resolution design follows AttnSleep [4] and is motivated physiologically: spindles at 11–16 Hz require sampling above 50 Hz, whereas the slow waves that dominate deep sleep are already captured at 25 Hz. Three consecutive epochs (588 tokens) form the input window. The model has 1.00 M parameters on the three-channel Sleep-EDF montage and 1.07 M on the seventeen-channel CHB-MIT montage, since the spatial convolution scales with the channel count.

The boundary module estimates changepoint probabilities from representation dissimilarity between tokens separated by offsets $s \in \{1, 4, 16\}$, spanning roughly 150 ms to 2.4 s. Writing $\hat{h}_s(t)$ for the ℓ_2 -normalized output of a per-scale two-layer projection,

$$d_s(t) = \sigma\left(-\frac{\tau}{2} \langle \hat{h}_s(t), \hat{h}_s(t+s) \rangle\right), \quad (1)$$

$$b(t) = \sigma(W_2 \text{GELU}(W_1 \mathbf{d}(t) + c_1) + c_2), \quad (2)$$

where σ is the sigmoid, $\mathbf{d}(t) = [d_1(t), d_4(t), d_{16}(t)]^\top$, $\tau > 0$ is a learned temperature shared across scales, and $W_1 \in \mathbb{R}^{6 \times 3}$, $W_2 \in \mathbb{R}^{1 \times 6}$. Equation (1) is a temperature-scaled cosine dissimilarity, chosen over Euclidean distance for scale invariance across electrodes and sessions, so d_s rises as adjacent representations diverge; the last s positions are compared against themselves. Equation (2) then fuses the three scales through a two-layer perceptron, allowing the model to weight short- and long-range dissimilarity differently rather than fixing their combination in advance. A consistency term $0.01 \sum_s \text{MSE}(d_s, \text{sg}[b])$, with sg the stop-gradient, holds the per-scale estimates near the fused output.

These probabilities condition attention through the cumulative boundary sum $C_i = \sum_{j=1}^i b(j)$, which rises wherever the model places boundary mass, and the same-regime affinity mask it induces,

$$S_{ij} = \exp(-|C_i - C_j|). \quad (3)$$

Since C grows only where boundaries are predicted, $|C_i - C_j|$ measures the boundary mass separating two tokens, so S_{ij} approaches one within a regime and decays across one. The eight attention heads are partitioned into three functional groups on this basis. Four *intra-regime* heads use S_{ij} and attend preferentially within detected

regimes, two *inter-regime* heads use $1 - S_{ij}$ to attend across boundaries, and two *cross-scale* heads stay unmasked for global context, giving $\tilde{a}_{ij}^{(h)} = a_{ij}^{(h)} + \log(M_{ij}^{(h)} + \epsilon)$ with $\epsilon = 10^{-6}$. The mask is recomputed from $\mathbf{b}$ inside each of the four blocks rather than once. After the final block, classification reads a mean over the 196 center-epoch tokens, with the flanking epochs acting as context.

3.2. ACBL with Gradient Isolation

The curriculum decouples the boundary head from the classification gradient while the boundary distribution takes shape. *Warmup* (epochs 0–2): classification loss only, letting the encoder develop representations the boundary head can fit. *Formation* (epochs 3–14): boundary probabilities are *detached* by a directional stop-gradient before entering the regime mask (Fig. 1), so the classifier cannot influence them and the shortcut of Section 2 is unavailable. *Full* (epochs 15+): boundaries reconnect, and the structure built during formation persists because the mask is by then content-dependent, so flattening the boundaries would now cost classification accuracy.

Four terms shape the boundary head. Let $\mathbf{p}$ be pseudo-targets from the min-max normalized cosine distance between adjacent stop-gradient encoder tokens, excluding epoch junctions, passed through a sigmoid with temperature annealed from 2.0 to 0.5 over 15 epochs; let t_j index the epoch junctions and y_j the epoch labels; and let $\bar{A}$ be the head-averaged attention of the final block, with P a row-normalized prior assigning 0.8 to same-stage token pairs and 0.2 otherwise. Then

$$\begin{aligned} \mathcal{L}_{\text{pseudo}} &= \text{BCE}(\mathbf{b}, \mathbf{p}), & \mathcal{L}_{\text{KL}} &= \text{KL}(P \parallel \bar{A}), \\ \mathcal{L}_{\text{Gauss}} &= \text{BCE}(\mathbf{b}, \mathbf{g}), & \mathcal{L}_{\text{var}} &= -\text{Var}_t(b(t)), \end{aligned} \quad (4)$$

where $g(t) \propto \sum_j \mathbf{1}[y_j \neq y_{j+1}] e^{-(t-t_j)^2/2s^2}$ with $s = 5$ tokens, $\mathbf{g}$ is normalized to unit peak, and $\mathcal{L}_{\text{Gauss}}$, $\mathcal{L}_{\text{KL}}$ are averaged over the windows in a batch containing at least one label transition. The variance term penalizes constant outputs directly. The full phase-dependent objective is

$$\mathcal{L}_{\text{total}} = \begin{cases} \mathcal{L}_{\text{CE}} & \text{Phase 1,} \\ \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{ACBL}}^{\text{detach}} & \text{Phase 2,} \\ \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{ACBL}} & \text{Phase 3,} \end{cases} \quad (5)$$

with $\mathcal{L}_{ACBL} = \alpha \mathcal{L}_{pseudo} + \beta (\mathcal{L}_{Gauss} + \gamma \mathcal{L}_{KL}) + \delta \mathcal{L}_{var}$ and $\alpha = 0.3$, $\beta = 0.5$, $\gamma = 0.5$, $\delta = 1.0$, $\lambda = 0.3$. The detach superscript denotes that $\nabla_{\theta_{bnd}} \mathcal{L}_{ACBL}$ is preserved while the classification gradient into θ_{bnd} is blocked. The asymmetry is essential: symmetric detachment leaves the boundary head decoupled from the encoder, while full coupling reproduces collapse (Section 4.2).

Phase lengths and weights were fixed a priori, reused unchanged across all three datasets and never selected on test data. A single-seed check over three (warmup, formation) settings, (3, 12), (1, 12) and (3, 16), gave accuracy 0.690–0.711 and boundary standard deviation 0.019–0.033, all above the null of Section 4.2, so the outcome is not knife-edge in the schedule even though the boundary magnitude is schedule-sensitive.

4. EXPERIMENTAL RESULTS

4.1. Datasets and Setup

We evaluate on three public clinical EEG datasets.

Sleep-EDF [1]: 147 subjects, three channels (two EEG, one electrooculogram), 5-class staging, subject-level 102/22/23 split, bandpass 0.5–35 Hz, in-bed periods only [4]. Hypnogram annotations in Sleep-EDF Expanded are variable-duration runs of one stage, and our preprocessing retained only annotations lasting exactly one 30-second epoch, giving 7,360 epochs biased toward short, transitional runs. Windows are assembled from index-adjacent retained epochs of the same subject without consulting recording time: only 33.5% of within-subject adjacent pairs are one epoch apart (median gap 180 s), and 160 of 1,025 test windows (15.6%) are contiguous in time. This affects how the boundaries and the accuracy comparison should be read (Sections 4.5, 4.6), but not the collapse results, which concern training dynamics.

CHB-MIT [14]: 12 subjects, 9,639 epochs, 17 bipolar channels, binary seizure detection (312 seizure, 9,327 normal), round-robin subject split balanced by seizure count, giving 1,496 test windows of which 53 are seizure-centered. Its preprocessing sweeps each recording contiguously.

TUAB [15]: 19 channels, binary normal/abnormal detection, random 60/20/20 split of 74,395 epochs. Since a window needs three consecutive epochs drawn from the same split, only 16,129/604/604 windows survive, and since the split is over epochs rather than recordings, training and test windows can originate from the same recording. TUAB therefore serves only as a specificity control for the boundary head; its classification numbers are not a cross-subject benchmark and are not compared with published results.

All experiments use AdamW (lr 10^{-4} , wd 10^{-4}), batch size 32 (16 on Sleep-EDF), gradient clipping 1.0, and up to 50 epochs, with early stopping (patience 12) enabled only after the formation phase, so that a run cannot terminate while the boundary distribution is still forming. Checkpointing and early stopping use the validation split alone, and every metric is computed on a held-out test split from the same validation-selected checkpoint, so classification and boundary statistics describe one model state. Headline results are mean $\pm$ standard deviation over five seeds (42–46) at a fixed split, isolating initialization and batch order. Tables 1 and 3 use a single seed, since each entry is a separate training run and the contrasts they isolate span an order of magnitude rather than turning on small differences; the conclusions drawn from them are qualitative, separating degenerate from non-degenerate outcomes. Deterministic kernels were not enforced, so runs vary slightly: two runs of the reported Sleep-EDF setting gave boundary standard deviations of 0.023 and 0.027, within the across-seed spread. Run-to-run variation of this size is therefore

the resolution at which the boundary statistics below should be read.

4.2. Auxiliary Head Collapse

When the boundary module is trained jointly with classification, it consistently converges to near-constant outputs (std ≤ 0.011), a failure we term *auxiliary head collapse*. The architectural cause is a *useful-when-flat* property of the same-regime mask: when $b(t)$ is constant, S_{ij} degenerates into an exponential distance-decay pattern that already supports above-0.70 accuracy, leaving no gradient incentive for the boundary head to differentiate. This is the shortcut pathology of Section 2.

Eleven controlled experiments across five categories probe this failure (Table 1). *Masking variants* (sigmoid, Gumbel-Softmax discretization, hard threshold) all collapsed to near-constant outputs with std ≤ 0.010 , so making the mask discrete does not alter the underlying dynamics. *ACBL loss variants* are cumulative: the boundary losses alone, then with the variance penalty added, then with the variance penalty and gradient isolation together. Only the last configuration is non-degenerate, which shows the variance penalty is necessary but far from sufficient, since penalizing constant outputs directly does not by itself create a gradient signal that distinguishes one boundary position from another. *Aggressive scheduling* (a bimodal loss with extended formation, and temperature annealing with focal loss) did not recover structure but instead found new degenerate solutions. *Architectural change* (cumulative product masking) collapsed to all-ones rather than all-zeros, so the masking operation sets the direction of the collapse while the pathology itself is invariant. *Alternative parameterizations* (a differentiable hidden Markov model [12] and query-based segmentation [13]) degenerated analogously despite representing boundaries in entirely different ways; their boundary statistic is not comparable with the others, so that judgment rests on accuracy together with inspection of the learned outputs.

A threshold on the boundary standard deviation means little without a reference for what an uninformative head produces. We therefore measured a null for the reported configuration by replacing each test window with a temporally shuffled surrogate, which destroys signal structure while preserving the epoch layout. Over 20 shuffles this gives boundary standard deviation 0.0154 (95th percentile 0.0155) against 0.026 ± 0.003 on real input, and all five seeds exceed it, by $1.47\text{--}1.94\times$. The 0.020 cutoff in Table 1 sits just above this null and is a descriptive label rather than a test. Two variants lie near it, bimodal scheduling just below at 0.014 and cumulative product just above at 0.018, so the statistic alone does not classify them. We treat both as degenerate on the strength of the failure observed during training, constant outputs and all-ones mask saturation respectively, together with their accuracy of 0.626 and 0.678 against 0.718 for gradient isolation, which at 0.028 is separated from them by margin as well as by kind. The null is measured for one model on one dataset and is not transferred to other variants or datasets.

4.3. Cross-Task Behaviour of the Boundary Head

Section 4.2 establishes that gradient isolation prevents collapse. Whether the boundaries it produces carry task content is a separate question, and the three datasets answer it differently (Table 2).

On Sleep-EDF the boundary distribution clears the shuffled-input null in every seed. CHB-MIT calls for a different test. Its boundary standard deviation is 0.014 ± 0.006 , but the null above was measured on the Sleep-EDF model and does not transfer across

Table 1. Eleven controlled interventions on Sleep-EDF, single seed. Bnd Std is the standard deviation of $b(t)$ pooled over all test windows, as in Table 2. Values below 0.020 label a degenerate boundary distribution; Section 4.2 calibrates this against a shuffled-input null.

Category	Experiment	Acc	Bnd Std
Masking	Sigmoid mask	0.703	0.010
	Gumbel-Softmax	0.642	0.003
	Hard threshold	0.677	0.004
ACBL loss	Boundary losses alone	0.667	0.011
	+ var. penalty	0.690	0.007
	+ grad. isolation	0.718	0.028
Scheduling	Bimodal + extended	0.626	0.014
	Temp. anneal + focal	0.666	0.006
Architectural	Cumulative product	0.678	0.018
Alt. param.	Differentiable HMM [12]	0.546	n/a
	Query-based [13]	0.707	n/a

datasets, so instead of comparing magnitudes we ask whether the boundary output responds to the presence of a transition at all. Splitting the 1,496 test windows into those containing a within-window label transition ($n=59$) and those without ($n=1,437$) shows no difference in pooled or per-window boundary standard deviation, with Mann-Whitney $p \geq 0.21$ in all five seeds and Cohen’s $d = -0.25 \pm 0.20$ pointing marginally the wrong way. A second test compares mean activation in the center epoch against the two flanking epochs. The center is higher in $98.9 \pm 1.7\%$ of seizure windows, which in isolation looks like a seizure effect, but it is also higher in $99.9 \pm 0.2\%$ of the 1,443 non-seizure windows, and the margin is in fact larger there in every seed ($d = -0.82 \pm 0.27$). The explanation is structural: the center epoch is bracketed by two epoch junctions while each flanking epoch borders only one, so a positive margin follows from the geometry of the window rather than from seizure content. The residual difference between the two groups is statistically clear but points away from the seizure hypothesis, so it cannot support that hypothesis at any effect size. Fig. 2 shows the effect directly: activation peaks at the fixed junctions where the Gaussian targets sit, whether or not a transition occurs there. We therefore report no learned boundary structure on CHB-MIT.

TUAB completes the picture from the opposite direction. Its labels apply to whole sessions, so no within-window transition exists by construction and the boundary output is correspondingly flat at 0.003, which is the outcome the task calls for. Taken alone this is weak evidence, since a collapsed head would look identical; what makes it informative is that the same configuration produces structure above the null on Sleep-EDF, so flatness here reflects the absence of transitions rather than a failure to learn. Turning to classification, the CHB-MIT κ of 0.356 ± 0.132 sits well below the AUROC of 0.924 ± 0.014 . This gap reflects severe class imbalance, at 3.5% seizure prevalence among test windows, which compresses chance-corrected agreement even when ranking performance is strong [16]; the wide spread of κ across seeds is a further symptom of the same imbalance, since a small change in the decision threshold moves few positives but many agreements.

The analyses above treat a boundary as correct if it falls at the right epoch. Sub-epoch resolution was assessed separately, by placing Gaussian targets at exact onsets taken from the seizure start and end times annotated in the CHB-MIT summary files. On a held-out subset ($n=7$) the boundary peak falls a median 0.46 s from the annotated onset, but this is a supervised alignment offset measured on

Table 2. Cross-task validation. Sleep-EDF and CHB-MIT are mean $\pm$ std over five training seeds at a fixed split; TUAB[†] is single-seed. Dashes mark metrics not defined for the task.

Dataset	Acc	AUROC	κ	Sens.	Bnd Std
Sleep-EDF	0.698 ± 0.013	—	0.596 ± 0.018	—	0.026 ± 0.003
CHB-MIT	0.901 ± 0.043	0.924 ± 0.014	0.356 ± 0.132	0.796 ± 0.047	0.014 ± 0.006
TUAB [†]	0.916	0.913	0.545	0.569	0.003

[†] Single seed; see Section 4.1 on the TUAB split. Its Bnd Std is a mean per-window standard deviation, not the pooled value used in the other rows.

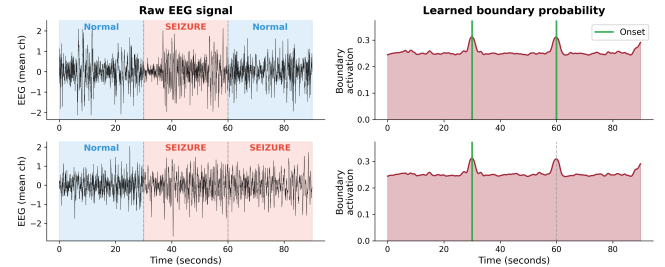


Fig. 2. Boundary activation on two CHB-MIT test windows. Each row is a 90-second window: EEG (left; normal blue, seizure red) with the learned boundary probability (right). Green lines mark annotated seizure onsets, grey dashed lines mark epoch junctions with no label change. Activation peaks at both junctions in both windows, including the junction in the lower row where the seizure continues and no transition occurs. The peaks track the fixed junctions at which Gaussian targets are placed rather than the annotated onsets.

too few cases to support a formal claim, and it is not a detection latency. Across all 53 seizure-centered test windows the nearest peak lies a median 30.0 s (mean 21.5 s) from the onset and within 5 s in 15 of them, whereas a PELT baseline [9] on the same windows reaches a median of 10.5 s. We therefore make no onset-localization claim. Event-detection latencies reported for continuous CHB-MIT systems [17] quantify the delay before an ongoing seizure is flagged, a different problem, and are not comparable with either figure.

4.4. Ablation Studies

Sections 4.2 and 4.3 establish when boundaries survive training and what they encode. Which part of ACBL is responsible is isolated by four ablations (Table 3), each removing one component. Removing the Gaussian targets causes complete collapse on both datasets. An externally anchored signal is therefore what stops the head drifting back to a degenerate optimum once gradients reconnect, and gradient isolation alone does not substitute for it: the curriculum protects the boundary head while it learns, but something must tell it what to learn. Gradient isolation matters most on the 5-class Sleep-EDF task (0.028 to 0.005) and less on binary seizure detection (0.035 to 0.030), where the stronger discriminative signal lets the ACBL losses compete with classification gradients even without detachment. Classification accuracy is preserved throughout, within 2% in every case, so none of these components is carrying the classifier.

4.5. Comparison with Published Methods

Table 4 places NeuroState alongside published methods. None of them provides structural boundary detection: AttnSleep and SleepTransformer expose post-hoc attention heatmaps only, L-

Table 3. Ablation of the ACBL components, single seed. Entries are boundary standard deviations averaged over batches, so they are comparable within this table but not with the pooled values of Tables 1 and 2.

Configuration	Sleep-EDF	CHB-MIT
Full model (all components)	0.028	0.035
No pseudo-boundary loss	0.005	0.019
No Gaussian boundary targets	0.0001	0.001
No attention KL prior	0.005	0.015
No gradient isolation	0.005	0.030

Table 4. Comparison with published methods.

Method	Acc	κ	Context	Interp.
<i>Sleep-EDF (5-class staging)</i>				
AttnSleep [4] [†]	0.829	0.77	3 ep.	Attn
SleepTransformer [5]	0.812	—	21 ep.	Attn+UQ
L-SeqSleepNet [6] [‡]	0.886	0.845	200 ep.	None
NeuroState [¶]	0.698	0.596	3 ep.	ACBL
<i>CHB-MIT (cross-subject seizure detection, AUROC)</i>				
BIOT vanilla [7] [§]	0.865	—	1 ep.	None
BIOT pre-trained [7] [§]	0.876	—	1 ep.	None
NeuroState	0.924	—	3 ep.	ACBL

[†]3-epoch result on Sleep-EDF-78 (Table VI of [4]). [‡]39 recordings with SHHS pretraining (Table III of [6]). [§]6/3/3 patient-wise split (Appendix B.2 of [7]); NeuroState uses a 6/3/3 round-robin split, both cross-subject.

[¶]147-subject subset with the epoch selection of Section 4.1.

SeqSleepNet offers no interpretability mechanism at all, and both BIOT configurations operate as black-box classifiers. The accuracy gap is therefore a deliberate trade, giving up staging accuracy for an auxiliary boundary output that none of these architectures produces, subject to the limits Sections 4.3 and 4.6 place on what that output has been shown to represent.

Several factors beyond architecture separate these numbers, and we set them out rather than attribute the gap to one cause. Our Sleep-EDF subset retains predominantly short, transitional stage runs (Section 4.1), the hardest epochs to classify, whereas the cited results evaluate full recordings dominated by stable epochs; subsets and protocols differ as well, as the footnotes record. Context length explains part of the remaining distance to the long-context models, since L-SeqSleepNet exploits whole-cycle structure over 200 epochs with Sleep Heart Health Study (SHHS) pretraining that a 3-epoch model cannot reach. It does not explain the distance to AttnSleep, which uses the same 3-epoch context, so that gap rests on the subset and the architecture. On CHB-MIT, AUROC 0.924 ± 0.014 stays above both BIOT configurations even one standard deviation below the mean, under broadly comparable patient-wise evaluation. Because the protocols are not identical, Table 4 reports published numbers together with the differences that bear on them rather than as a matched comparison.

4.6. Concept-Based and Spectral Analyses

The preceding sections describe the boundary head through summary statistics. This one asks a different question: what do the learned representations encode, and do the boundaries fall where physiology would predict? Applying TCAV [18] with concepts defined by band power (delta, spindle, alpha; concept classifiers at

0.84–0.89 cross-validated accuracy), the patterns match clinical expectation: alpha drives Wake (0.81), the spindle band drives N1 (0.75), and high-frequency (0.81) and high-amplitude (0.77) content drive seizure detection. Splitting tokens into high- and low-boundary quintiles, Sleep-EDF high-boundary regions show $2.1\times$ higher beta and $3.3\times$ higher gamma power ($p < 10^{-22}$), and CHB-MIT high-boundary regions $\sim 25\%$ lower amplitude variability ($p < 10^{-38}$), both surviving Bonferroni correction.

Taken alone, these results would suggest that the boundary head has learned physiologically meaningful structure. We present them as preliminary instead, for three reasons. The Gaussian targets are placed at known label transitions, so validating the boundaries against those same transitions is partly circular. On Sleep-EDF most high-boundary tokens sit at epoch junctions that are not contiguous in recording time (Section 4.1), so the spectral contrast may reflect different times of night rather than transition physiology; consistent with this, the boundary standard deviation is indistinguishable between contiguous and spliced windows, at 0.0221 against 0.0226. On CHB-MIT, Section 4.3 finds no transition-related structure to interpret. Establishing physiological meaning would require contiguous preprocessing together with boundary supervision independent of the labels used to validate it. The concept results stand on their own as evidence about the encoder representations; it is the step from those representations to the boundary positions that remains unsupported.

5. CONCLUSION

This paper treats structurally interpretable boundary detection inside an EEG transformer as an optimization problem rather than an architectural one. Auxiliary head collapse proves to be a robust failure mode, defeating masking variants, loss reformulations, aggressive schedules and alternative parameterizations alike, and gradient isolation is the only remedy among the eleven interventions tested. That a single mechanism resolves the collapse across parameterizations as different as sigmoid masks, discrete masks, hidden Markov models and segmentation queries indicates that the pathology lies in the joint optimization itself rather than in any particular representation of the boundary. The practical implication is that an auxiliary head which appears to train successfully, in the sense that the total loss falls and the primary task improves, may nonetheless have learned nothing, and that detecting this requires a statistic on the auxiliary output together with a null to compare it against. We scope the conclusion to the EEG settings measured here.

Interpreting the boundaries is a further step, and the evidence remains preliminary. On Sleep-EDF they clear a shuffled-input null, yet respond alike to genuine transitions and to joins between distant epochs; on CHB-MIT they do not distinguish transition-containing windows from the rest; on TUAB they are flat, as the task requires. Sub-epoch onset localization is likewise not established. The optimization result and its interpretation are therefore separable, and the former does not depend on the latter.

Future work includes re-preprocessing Sleep-EDF so that every window is contiguous in recording time, splitting TUAB at the recording level, adding matched-protocol baselines, extending the intervention sweep to multiple seeds, and supervising the boundary head independently of the labels used to validate it. Applying the same curriculum to auxiliary heads outside EEG would test how far the optimization result reaches.

6. REFERENCES

- [1] B. Kemp, A. H. Zwinderman, B. Tuk, H. A. C. Kamphuisen, and J. J. L. Obery, “Analysis of a sleep-dependent neuronal feedback loop: the slow-wave microcontinuity of the EEG,” *IEEE Trans. Biomed. Eng.*, vol. 47, no. 9, pp. 1185–1194, 2000.
- [2] F. Mormann, R. G. Andrzejak, C. E. Elger, and K. Lehnertz, “Seizure prediction: the long and winding road,” *Brain*, vol. 130, no. 2, pp. 314–333, 2007.
- [3] C. Iber, S. Ancoli-Israel, A. L. Chesson, and S. F. Quan, *The AASM Manual for the Scoring of Sleep and Associated Events*, American Academy of Sleep Medicine, 2007.
- [4] E. Eldele *et al.*, “An attention-based deep learning approach for sleep stage classification with single-channel EEG,” *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 29, pp. 809–818, 2021.
- [5] H. Phan *et al.*, “SleepTransformer: Automatic sleep staging with interpretability and uncertainty quantification,” *IEEE Trans. Biomed. Eng.*, vol. 69, no. 8, pp. 2456–2467, 2022.
- [6] H. Phan *et al.*, “L-SeqSleepNet: Whole-cycle long sequence modelling for automatic sleep staging,” *IEEE J. Biomed. Health Inform.*, vol. 27, no. 10, pp. 4748–4757, 2023.
- [7] C. Yang, M. B. Westover, and J. Sun, “BIOT: Biosignal transformer for cross-data learning in the wild,” in *NeurIPS*, 2023.
- [8] W.-B. Jiang, L.-M. Zhao, and B.-L. Lu, “Large brain model for learning generic representations with tremendous EEG data in BCI,” in *ICLR*, 2024.
- [9] R. Killick, P. Fearnhead, and I. A. Eckley, “Optimal detection of changepoints with a linear computational cost,” *J. Amer. Statist. Assoc.*, vol. 107, no. 500, pp. 1590–1598, 2012.
- [10] Y. Ishikawa, S. Kasai, Y. Aoki, and H. Kataoka, “Alleviating over-segmentation errors by detecting action boundaries,” in *WACV*, 2021, pp. 2322–2331.
- [11] R. Geirhos *et al.*, “Shortcut learning in deep neural networks,” *Nat. Mach. Intell.*, vol. 2, no. 11, pp. 665–673, 2020.
- [12] K. M. Tran, Y. Bisk, A. Vaswani, D. Marcu, and K. Knight, “Unsupervised neural hidden Markov models,” in *EMNLP Workshop on Struct. Predict. NLP*, 2016, pp. 63–71.
- [13] B. Cheng, I. Misra, A. G. Schwing, A. Kirillov, and R. Girshick, “Masked-attention mask transformer for universal image segmentation,” in *CVPR*, 2022, pp. 1290–1299.
- [14] A. H. Shoeb and J. V. Gutttag, “Application of machine learning to epileptic seizure detection,” in *ICML*, 2010, pp. 975–982.
- [15] I. Obeid and J. Picone, “The Temple University Hospital EEG data corpus,” *Front. Neurosci.*, vol. 10, p. 196, 2016.
- [16] A. R. Feinstein and D. V. Cicchetti, “High agreement but low kappa: I. The problems of two paradoxes,” *J. Clin. Epidemiol.*, vol. 43, no. 6, pp. 543–549, 1990.
- [17] Y. G. Chung, A. Cho, H. Kim, and K. J. Kim, “Single-channel seizure detection with clinical confirmation of seizure locations using CHB-MIT dataset,” *Front. Neurol.*, vol. 15, art. 1389731, 2024.
- [18] B. Kim *et al.*, “Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (TCAV),” in *ICML*, 2018, pp. 2668–2677.